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SUMMARY

The atoms were believed to be non -divisible until the discovery of subatomic particles. J. J. Thomson proposed that the atom is a positively charged sphere in which the electrons are embedded. However, it fails to explain the stability of atoms. Rutherford, based on his α-rays scattering experiment, introduced the term nucleus which is a positively charged one and the negatively charged electrons are revolving around it, at high speeds. Bohr modified the Rutherford theory and introduced stationary orbits by taking into account the quantisation of energy. Louis de Broglie proposed that all matter possess dual nature. i.e. they behave both as a wave and a particle. De Broglie wavelength $\lambda = h / mv = h / \sqrt{2mev}$ is significant for a microscopic particle such as an electron. The wave nature of electron is also proved by Davisson and Germer through electron diffraction. For a microscopic particle such as an electron, the simultaneous measurement of the conjugate variables such as position and momentum involves uncertainty which is known as Heisenberg's uncertainty principle and it is expressed as
$\Delta x.\Delta p \geq h/4\pi$.

De Broglie's concept and Heisenberg's uncertainty principle lead to the development of quantum mechanical model of atom. Erwin Schrodinger, developed an equation for an electron wave which is expressed as $H\psi = E\psi$. This second order differential equation is exactly solvable for one electron system such as H, He^+ etc... but it is quite complex for multi-electron systems. The Schrödinger wave equation is solvable for certain energy values called eigen values. The wave functions corresponding to these eigen values are called atomic orbitals. The wave function ψ itself has no physical meaning. However, $|\psi|^2$ is related to the probability of finding electron around the nucleus. Thus the quantum mechanical model introduced us the term orbital which is the three dimensional space in which the probability of finding the electron is maximum. The electron in an orbital can be described by a set of four quantum numbers namely, principal quantum number (n) representing the principal energy level, azimuthal quantum number (l) representing the sub shell (orbital), magnetic quantum number (m) representing the different orientation of orbitals in space and spin quantum number (s) representing the spinning of electron about its own axis.

The general solution of Schrodinger for a one electron system can be expressed in spherical polar coordinates (r, θ, φ)

$$\Psi (r, \theta, \varphi) = R(r).f(\theta).g(\varphi)$$

(where R(r) is called radial wave function, other two functions are called angular wave functions).

The plot of $4\pi r^2\psi^2$ versus r gives the radial distribution curves. The number of radial nodes is given by$(n-l-1)$ whereas the angular nodes is equal to l. The angular distribution curve gives the boundary space diagram of orbital. s